

TURN 5: EXACT CHANNEL WINDOWS, PRIMITIVE PAIR FIBRES, AND RECIPROCAL-SOURCE OBSTRUCTIONS IN ERDŐS–STRAUS

THE CLANKERS

ABSTRACT. An original divisor involution interchanges the exterior and middle channels. For a source $a = 2^e A$ the associated windows join to length $3e + 3$; with one simple nonresidue prime this gives complete channel counts and a sharp one-unit comparison. A common p -coloured pair source has exact fibres $2\tau(h)$ and gives the complete first-half count after Möbius inversion. Its factor-source Jacobi term is annihilated. Reciprocal-source and six-vertex examples retain empty original coefficients despite actual factor supply; neither is an ES counterexample. All maps and finite checks retain their original exponent bounds, channel labels and ordered returns.

1. TURN 5: WHAT HAS AND HAS NOT BEEN FORCED

The preceding square-source theorem supplies actual new nonresidue factors at a prescribed hard prime, but does not imply that their original exterior or middle coefficient is occupied. This tranche gives a precise channel-separation theorem, a common integer pair source with its complete fibres, and a reciprocal-source comparison. A direct original-divisor involution and sharp joined windows provide a positive channel coupling; a complete six-vertex example records its precise limits within the expanded source system. It also gives a finite counterexample to the proposal that appending the reciprocal source rescues every locally empty square source. None of these results is a proof or counterexample to Erdős–Straus (ES).

The Type I/II parametrizations and the reciprocal master equation are antecedents [1, §2][2, §§1–4]. They are rederived below in the present marking. No historical-priority claim is made. The present arguments distinguish the exact arithmetic obstruction, the fibre-normalized count, and the missing universal lower bound. The recovered interrupted Turn 5 draft has been rechecked and is extended by the channel-window and six-vertex results below; no earlier execution record is inferred from it.

2. ORIGINAL SOURCE AND A CHANNEL-SEPARATION THEOREM

Let $p \equiv 1 \pmod{4}$ be prime, $p/4 < a < p$, $R = 4a - p > 0$, and $\gcd(pa, R) = 1$. The first-half atlas additionally requires $a < p/2$; square-source auxiliary vertices can lie above that bound. Factor $a = \prod_{\ell} \ell^{e_{\ell}}$. The original coordinates are

$$f_{\ell} \in [0, 2e_{\ell}]_{\mathbb{Z}}, \quad u = \prod_{\ell} \ell^{f_{\ell}} \mid a^2, \quad \beta_{\ell} = f_{\ell} - e_{\ell}. \quad (1)$$

Both maps are bijective with their displayed prime-valuation inverses. The channels are

$$E : R \mid 4u + 1, \quad M : R \mid 4u + p \iff R \mid u + a. \quad (2)$$

With $d = \gcd(a, u)$ put $h = d^2/u$, $r = u/d$, $s = a/d$. Prime valuations give $a = hrs$, $u = hr^2$, $\gcd(r, s) = 1$ and positivity. The returns are

$$\begin{aligned} E : \kappa &= (pr + s)/R, & (x, y, z) &= (a, hsk, phr\kappa), \\ M : \lambda &= (r + s)/R, & (x, y, z) &= (a, phs\lambda, phr\lambda). \end{aligned} \quad (3)$$

The gates imply the integer quotients; multiplication verifies $4/p$. The ordered inverses are $u = a^2/(Ry - pa)$ and $u = pa^2/(Ry - pa)$ respectively. These identities are inherited [1, Propositions 2.2 and 2.6].

Write χ_p for the quadratic character modulo p and $(\cdot/R)$ for the Jacobi symbol, including composite R .

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Theorem 2.1 (Exact character separation). *For every original divisor u ,*

$$(u/R) = \chi_p(u).$$

A genuine E hit has $\chi_p(u) = \chi_p(h) = -1$. A genuine M hit has

$$\chi_p(u) = \chi_p(h) = -\chi_p(a). \quad (4)$$

In particular, on every negative source $\chi_p(a) = -1$, E requires a nonresidue original divisor and M requires a residue original divisor.

Proof. For an odd prime $\ell \mid a$, reciprocity and $R \equiv -p \pmod{\ell}$ give

$$(\ell/R) = (-1/\ell)(R/\ell) = (-1/\ell)(-p/\ell) = (p/\ell) = (\ell/p).$$

If $2 \mid a$, then $R \equiv -p \pmod{8}$, and the supplementary law gives the same identity for 2. Multiplication proves the assertion for all original u . In E, $u \equiv -4^{-1} \pmod{R}$, whose Jacobi symbol is -1 since $R \equiv 3 \pmod{4}$. In M, $u \equiv -p4^{-1}$, giving $-(p/R)$. But $(p/R) = (a/R) = \chi_p(a)$. Finally $u = hr^2$ and $p \nmid r$ give the assertions about h . $\square$

The reciprocity input is classical [3, §6.4]. No sign is inferred from a positive real value or from the name of a channel.

For actual square sources $a = (p + \sigma_q q t^2)/4 < p$, with q a nonresidue prime, $\sigma_q \in \{1, 3\}$ as in Turn 4 and p hard, $\chi_p(a) = -1$. Consequently a common- u map between two such sources cannot carry an E hit to an M hit. More generally a rational square ratio u'/u , prime to p , preserves the character and cannot cross the channels. A crossing must change an odd total exponent on nonresidue prime coordinates. This is a necessary condition, not a sufficient target test.

The negative-source restriction is essential. On an arbitrary original shell,

$$E_a \cap M_a = \emptyset \quad \text{if } R \nmid p-1, \quad E_a = M_a \quad \text{if } R \mid p-1. \quad (4a)$$

This follows by subtracting the two gate numerators. In the second case $\chi_p(a) = (p/R) = 1$. Equality of the two gate sets does not assert that either is nonempty. At the hard prime $p = 1009$, however, $R = 3, a = 253, u = 11$ is in both. Its two tagged returns are $(253, 87032, 3818056)$ and $(253, 2042216, 88792)$. The two states must not be merged because their original divisor happens to coincide. The complete-shell target retains positive- as well as negative-character sources; the nonresidue-factor graph alone does not.

The complete available character counts, before any gate, are

$$N_{\pm}(a) = \frac{1}{2} \left(\prod_{\ell} (2e_{\ell} + 1) \pm \prod_{\chi_p(\ell)=1} (2e_{\ell} + 1) \right). \quad (5)$$

Indeed a nonresidue factor contributes $\sum_{f=0}^{2e} (-1)^f = 1$ to the signed count, while a residue factor contributes $2e + 1$. Thus both sign inventories can be nonempty while both target coefficients vanish.

Exact bounded crossing at one source. For a specified rational ratio $\rho = \prod \ell^{\delta_{\ell}}$ supported on the original prime set, the map $f \mapsto f + \delta$ has exactly the domain

$$\max(0, -\delta_{\ell}) \leq f_{\ell} \leq \min(2e_{\ell}, 2e_{\ell} - \delta_{\ell}) \quad \text{for every } \ell. \quad (6)$$

Its inverse subtracts δ . On this domain E maps to M if $\rho \equiv p \pmod{R}$; M maps to E if $\rho \equiv p^{-1} \pmod{R}$. Conversely either channel crossing forces that residue equation, since the original u is a unit modulo R . These are exact domains and maps, not an assertion that such a ratio is always available.

3. A DIRECT CROSSING AND THE EXACT JOINED ARITHMETIC WINDOWS

The preceding character obstruction specifies what a successful crossing must change. There is also a concrete positive crossing, and it extends to a sharp joint window on an original factor packet. None of the following constructions adds a factor that is absent from the source integer.

Theorem 3.1 (The maximal divisor domain of a channel involution). *At an original source with $4 \mid a$, put $V = \text{Div}(a/4)$. The map*

$$\mathcal{T}(u) = \frac{a}{4u} \quad (u \in V) \quad (\text{W1})$$

is an involution of V and interchanges the tagged E and M hit sets on V . Among all original divisors $u \mid a^2$, this is its complete integral return domain. Retaining the channel tag, its inverse is the same formula. Its only possible numerical fixed points satisfy $a = 4u^2$; they are not identified across the two channel labels.

Proof. $\mathcal{T}(u)$ is an integer if and only if $4u \mid a$. In that case it and u divide $a/4$, and the second application returns u . Every such divisor is a unit modulo R . Thus

$$R \mid 4u + 1 \iff R \mid a(4u + 1) = 4u(\mathcal{T}(u) + a) \iff R \mid \mathcal{T}(u) + a.$$

The character ratio is $\chi_p(\mathcal{T}(u)/u) = \chi_p(a)$ because $\mathcal{T}(u)/u = a/(4u^2)$. Thus this is the actual non-square crossing required on a negative source, rather than a forbidden square-ratio transport. This also proves the reverse channel assertion. The fixed-point equation is $4u^2 = a$. In particular negative-character sources have no numerical fixed point. The positive-source example $p = 61, a = 16, R = 3, u = 2$ is a fixed point, with two different tagged denominator returns. $\square$

Write $a = 2^e A$, A odd, and suppose $e \geq 1$. Retain only the specified packet

$$\mathcal{D} = \{2^f d : 0 \leq f \leq 2e, d \mid A\} \subseteq \text{Div}(a^2). \quad (\text{W2})$$

The omitted original divisors, whose odd parts divide A^2 but not A , remain outside this packet, not outside the complete atlas. For each $w \mid A$ define

$$I_E = [-2e - 2, -2]_{\mathbb{Z}}, \quad I_M = [-e, e]_{\mathbb{Z}}, \quad 2^j \equiv -w \pmod{R}. \quad (\text{W3})$$

Theorem 3.2 (Joined-window coefficient theorem). *The maps*

$$(w, j, E) \mapsto u_E = 2^{-j-2}w, \quad (w, j, M) \mapsto u_M = 2^{e+j}A/w \quad (\text{W4})$$

are bijections from the respective solutions of (W3) in I_E, I_M onto the original tagged packet hits. Their inverses read $f = v_2(u)$, the odd part d , and return $(w, j) = (d, -f - 2)$ for E and $(A/d, f - e)$ for M . The union of the two integer windows is the interval

$$[-2e - 2, e]_{\mathbb{Z}}, \quad |I_E \cup I_M| = 3e + 3. \quad (\text{W5})$$

On their intersection $[-e, -2]$ the two divisors have product $a/4$, giving exactly the crossing (W1). For $e = 1$ that intersection is empty.

Let $d_R = \text{ord}_R(2)$ and $A_- = \#\{w \mid A : -w \in \langle 2 \rangle_R\}$. If $3e + 3 \geq d_R$, the packet is jointly occupied if and only if $A_- > 0$. Moreover put

$$2e + 1 = f_0 d_R + \rho, \quad 0 \leq \rho < d_R, \quad \Delta = (e + 2) \bmod d_R, \quad \delta = \min(\Delta, d_R - \Delta).$$

Then its exact joint count C_{packet} obeys

$$A_-(2f_0 + \mathbf{1}_{\rho \geq d_R - \delta}) \leq C_{\text{packet}} \leq A_-(2f_0 + \mathbf{1}_{\rho > 0} + \mathbf{1}_{\rho > \delta}). \quad (\text{W6})$$

The bracketed constants are the minimum and maximum of the paired cyclic count at a target class. In particular every cyclic target is covered exactly when $3e + 3 \geq d_R$; this is an exact interval-cover threshold, not an assertion that every abstract target is realized by a prime source.

Proof. For E , $4u + 1 = 0$ is equivalent to $2^{-f-2} = -d$. For M , divide $2^f d + 2^e A = 0$ by the unit $2^e d$ to obtain $2^{f-e} = -A/d$. This proves (W3)–(W4) and their inverses. Their integer intervals have no gap when $e \geq 1$. On the overlap the displayed product is $2^{e-2}A = a/4$.

For the count, remove f_0 complete periods from each window. The two remaining cyclic intervals have length ρ and displacement Δ . They overlap precisely when $\rho > \delta$, and cover the whole cycle precisely when $\rho \geq d_R - \delta$. Thus the minimum of their indicator sum is zero or one as in (W6), and its maximum is zero, one or two as stated. Sum this exact bound over the A_- actual divisors; a divisor outside the subgroup contributes zero. For coverage, (W5) contains a complete period precisely at its stated length. $\square$

The two label occurrences in an overlap are not one state. For $e = 0$ the windows are the singleton sets $\{-2\}$ and $\{0\}$; (W5)–(W6) are not used. The same prime's upper-half sources remain in the data; their M packet is empty by the already proved denominator-size argument.

4. A COMPLETE CLASSIFICATION WHEN THE NONRESIDUE FACTOR IS SIMPLE

The joined windows admit a complete rather than subpacket interpretation on a specified original factor class. This separates an actual odd-component hole from merely insufficient bounded powers.

Theorem 4.1 (One simple nonresidue factor). *Suppose $a = Br$, r is prime, $r \nmid B$, $(r/p) = -1$, and every prime factor of B is a residue modulo p . Let*

$$\mu_B(g) = \#\{v \mid B^2 : vB^{-1} = g \pmod{R}\}.$$

Then the complete original channel counts are

$$\boxed{|E_a| = \mu_B(-p^{-1}), \quad |M_a|/2 = \mu_B(-r).} \quad (\text{W7})$$

The exact E fibre is $u = rv$, with $vB^{-1} = -p^{-1}$. The first M orientation is $u = v$, with $vB^{-1} = -r$, and its paired orientation is a^2/v . The complements are distinct and exhaust M. All original exponent vectors of B and their fine residue multiplicities are retained.

Proof. The original source has character -1 . The channel-character theorem forces an odd exponent of r for E and an even exponent for M. The only permitted exponents are 0, 1, 2. At exponent one, $u/a = v/B$, giving the E formula. At exponent zero, $u/a = v/(Br)$, so M requires $v/B = -r$. Complementation sends this orientation to exponent two and is a bijection, proving the full count. $\square$

In particular let $B = \ell^e$, with ℓ prime and $(\ell/p) = 1$. Put $H = \langle \ell \rangle_R$, $d_R = |H|$. If $4 \in H$, retain $c = \log_\ell 4 \in [0, d_R]$; if $-r \in H$, retain $j = \log_\ell(-r)$. The complete counts become

$$\boxed{\begin{aligned} |E_a| &= \#\{j' \in [-2e - c, -c] : j' \equiv j \pmod{d_R}\}, \\ |M_a|/2 &= \#\{j' \in [-e, e] : j' \equiv j \pmod{d_R}\}. \end{aligned}} \quad (\text{W8})$$

If $-r \notin H$, both vanish. The E map is $u = \ell^{-j'-c}r$, and the first M map is $u = \ell^{e+j'}$, with the complementary M word retained. Each window has length $2e + 1$, hence

$$\boxed{||E_a| - |M_a|/2| \leq 1.} \quad (\text{W9})$$

If $4 \notin H$, M requires $-r \in H$, while E requires $-r \in 4^{-1}H$. Those are disjoint cosets, so at most one channel can occur. No nonexistent logarithm c is assigned in that case.

For $\ell = 2$, one may use $c = 2$ as an integer lift regardless of the order, and Theorem 3.2 is an exact classification of $|E| + |M|/2$ on this factor class, not just a subpacket. The full M count is twice its first orientation count. At fixed R and fixed odd factor A , increasing e by d_R adds two occurrences per target to each labelled packet in (W3). This is an all-exponent recurrence with $p_e = 4 \cdot 2^e A - R$ varying; primality and the original source range must be checked for each parameter.

The cyclic threshold has an actual sharp boundary example:

$$p = 23209, \quad R = 127, \quad a = 2 \cdot 2917, \quad d_R = 7, \quad -r \equiv 2^2 \pmod{127}.$$

The E window occupies 3, 4, 5, the M window 6, 0, 1, and class 2 is absent. Both complete channels are empty one class below $3e + 3 \geq d_R$. At the hard primes 43801, 8521, 12601, with residual 31 and respectively $a = 2 \cdot 5479, 2 \cdot 1069, 2 \cdot 1579$, the complete pairs $(|E|, |M|/2)$ are (1, 0), (0, 1), (1, 1). The one-unit difference in (W9) cannot be removed. These primes and every factor are checked by trial division.

5. SIX EXPANDED VERTICES CAN STILL HAVE EMPTY COMPLETE SQUARE-SOURCE CHANNELS

For a prime vertex q , write $\sigma_q = 1$ at $q \equiv 3 \pmod{4}$ and $\sigma_q = 3$ at $q \equiv 1 \pmod{4}$. Its square source has $R = \sigma_q q t^2$, $a = (p + R)/4$, and positive odd t with $R < 3p$. Turn 4 forces six distinct reachable vertices, not a selector. Here is an exact local obstruction at that cardinality, beyond the preceding two-vertex example.

Theorem 5.1 (A complete six-vertex local failure certificate). *At the hard prime $p = 315361 \equiv 361 \pmod{840}$, the canonical graph has the closed six-cycle*

$$27847 \rightarrow 42901 \rightarrow 13877 \rightarrow 2789 \rightarrow 20233 \rightarrow 18803 \rightarrow 27847. \quad (\text{W10})$$

Every vertex has exactly its displayed outgoing nonresidue prime, to exponent one. At all 17 valid square sources of these six vertices, both original E/M channels are empty. The complete source consists of 663 original divisors. This is not a closed set in the expanded square graph and is not an ES counterexample.

Proof. The six complete canonical factorizations are

$$2 \cdot 42901, \quad 2^3 \cdot 13877, \quad 2^5 \cdot 2789, \quad 2^2 \cdot 20233, \quad 5 \cdot 18803, \quad 3 \cdot 27847.$$

Exact trial division proves p and all factor primes. Their quadratic symbols prove the stated complete successor set. The finite t-ranges, in that order, are $\{1, 3, 5\}$, $\{1\}$, $\{1, 3\}$, $\{1, 3, 5, 7, 9\}$, $\{1, 3\}$, and $\{1, 3, 5, 7\}$. They follow from the strict inequality $\sigma_q q t^2 < 3p$. The certificate lists every factorization and every bounded divisor and checks both gates directly. Every source in fact has a unique simple nonresidue factor, so (W7) independently reduces the two full gates to the two exact positive-factor coefficients. Both reductions are checked against the complete divisor list.

For the canonical source $q = 42901$, the pure-2 subgroup modulo 128703 has order 14300 and contains neither required target. Several other sources have their target in the generated subgroup but outside their actual bounded packet. These are two different certified reasons for vanishing. $\square$

An actual exit and successful next source are retained. At $q = 20233, t = 3$,

$$A = 215413 = 11 \cdot 19583$$

has the nonresidue factor 11. The canonical source at that prime is $A_{11} = 78843 = 3 \cdot 41 \cdot 641$ and has an exterior hit $u = 41$ as well as four middle states. Its E normalization and ordered triple are in the certificate. This is an actual path, not a proof that every newly produced factor reaches a hit. No leastness or complete-square-graph failure is asserted for (W10).

The finite certificate is independent of the discovery search that suggested this cycle. The proof only needs the displayed prime, its six complete factor identities, the exact valid source ranges, and all 663 divisor vectors. The independent checker uses direct integer divisors and literal square-residue sets, not the first implementation's signed-box recurrence or Euler-symbol routine.

6. ONE COMMON INTEGER PAIR SOURCE FOR THE TWO CHANNELS

For this section $R \geq 3$ is odd and $N \geq 1$ has $\gcd(pN, R) = 1$ and $p \nmid N$; N need not be an actual distinguished denominator. Define

$$\mathcal{P}_R(N) = \{(b, c) : b, c \mid pN, R \mid b + c\}, \quad T_R(N) = |\mathcal{P}_R(N)|. \quad (7)$$

Let $\mathcal{F}_R(N)$ be the disjoint union of ordered coprime pairs (r, s) , $rs \mid N$, with labels

$$M : R \mid r + s, \quad E : R \mid pr + s,$$

and let $F_R(N)$ denote its cardinality. Only at $N = a$, $4a = p + R$, is this the original ES state set. At auxiliary N the label records the congruence, not a spurious ES solution with $4N = p + R$.

Theorem 6.1 (Exact pair fibres). *Every element of $\mathcal{F}_R(N)$, with $h = N/(rs)$, has exactly $2\tau_{\text{div}}(h)$ preimages in $\mathcal{P}_R(N)$. The complete inverse fibres are*

$$M : (b, c) = (p^\varepsilon dr, p^\varepsilon ds), \quad \varepsilon \in \{0, 1\}, \quad d \mid h; \quad (8)$$

$$E : (b, c) = (pdr, ds) \text{ or } (ds, pdr), \quad d \mid h. \quad (9)$$

In particular

$$T_R(N) = 2 \sum_{(c,r,s) \in \mathcal{F}_R(N)} \tau_{\text{div}}(N/(rs)), \quad F_R(N) = \sum_{(b,c) \in \mathcal{P}_R(N)} \frac{1}{2\tau_{\text{div}}(h(b,c))}. \quad (10)$$

Proof. Write $g = \gcd(b, c)$, $x = b/g$, $y = c/g$. Since $b, c \mid pN$ and $p \nmid N$, the exponent of p in each is at most one. If p divides neither x nor y , put $(r, s) = (x, y)$ and write $g = p^\varepsilon d$. The original lcm condition $\text{lcm}(b, c) \mid pN$ gives $drs \mid N$, and $R \mid b + c$ gives $R \mid r + s$ because p, d are units. This is precisely (8). If exactly one of x, y contains p , remove that one factor and designate the result r and the other side s . Now $g = d$, $drs \mid N$, and $R \mid pr + s$. This gives (9), including its two distinct orderings. They cannot overlap with (8), which has equal p -valuations on its two coordinates. All reconstruction maps are literal inverses. At an actual shell, $u = hr^2$ is the original divisor and (3) returns its denominators. $\square$

Here $\tau_{\text{div}}(n) = |\text{Div}(n)|$ is the divisor-count function; it is not the absent-support symbol τ of SplitZero. Likewise μ_M below denotes the Möbius function, not the original residue-count array.

The common-prime bit in M and the side-of- p bit in E are different markings. They are not discarded when taking a greatest common divisor.

Put

$$B_0 = \#\{b, c \mid N : R \mid b + c\}, \quad B_1 = \#\{b, c \mid N : R \mid b + pc\}.$$

The four p -colours have matrix

$$\begin{pmatrix} B_0 & B_1 \\ B_1 & B_0 \end{pmatrix}, \quad T_R(N) = 2(B_0 + B_1). \quad (11)$$

At an actual shell $B_0 = \sum_M \tau_{\text{div}}(h)$ and $B_1 = \sum_E \tau_{\text{div}}(h)$. The matrix need not be positive semidefinite: positive integer entries do not imply that its difference eigenvalue $B_0 - B_1$ is nonnegative.

7. MÖBIUS REMOVAL OF THE EXACT COMMON-DIVISOR MULTIPLICITY

Theorem 7.1 (Complete primitive count). *With (7),*

$$\frac{1}{2}T_R(N) = \sum_{d \mid N} F_R(N/d), \quad \boxed{F_R(N) = \frac{1}{2} \sum_{d \mid N} \mu_M(d) T_R(N/d)}. \quad (12)$$

At an actual first-half shell $F_R(a) = |E_a| + |M_a|$; consequently the complete primewise count is exactly

$$\boxed{\mathcal{C}_p = \frac{1}{2} \sum_{p/4 < a < p/2} \sum_{d \mid a} \mu_M(d) T_{4a-p}(a/d)}. \quad (13)$$

Here $\mathcal{C}_p = \sum_a (|E_a| + |M_a|)$ counts tagged ordered states. The distinct increasing-triple count is $\sum_a (|E_a| + |M_a|/2)$; the two counts are not equated. ES at p is equivalent to $\mathcal{C}_p > 0$, using the first-half completeness theorem retained in Turns 1–2. Equation (13) does not establish that inequality.

Proof. For a primitive pair with $rs \mid N$, each common divisor $d \mid N/(rs)$ is one term of $F_R(N/d)$. This proves the first equality with all multiplicities. Summing it against μ gives a coefficient $\sum_{d \mid m} \mu_M(d) = \prod_{q \mid m} (1 - 1)$, equal to one for $m = 1$ and zero otherwise. This proves the second identity. The maps $u \leftrightarrow (h, r, s)$ prove the actual-shell identification. The auxiliary pairs use only divisors of $p(a/d)$, nested in the original pa ; they are not relabelled as shells at a/d . $\square$

The underlying operator has an exact original metric interpretation. Let $f_N(g)$ count the original divisors of N in residue g , use the counting inner product on $\mathbb{R}^G$, and define the coefficient permutations

$$(S_p f)(g) = f(p^{-1}g), \quad (J_- f)(g) = f(-g), \quad C_p = I + S_p.$$

Then

$$\boxed{T_R(N) = \langle C_p f_N, J_- C_p f_N \rangle}. \quad (14a)$$

The matrix (11) is the pullback of this *signed* form to the two columns $f_N, S_p f_N$, not an ordinary positive Gram. Its kernel comparison is literal: $C_p f = 0$ exactly when $f(pg) = -f(g)$ for all g . On nonnegative divisor counts $C_p f_N$ is nonzero; a vanishing character observation of it is a supported zero, not absence of the original divisor source. Replacing J_- by the identity would calculate a different coefficient and cannot prove the desired target.

For existence alone the Möbius subtraction is unnecessary. At an actual shell, $h \mid a$, so the exact fibres prove

$$2F_R(a) \leq T_R(a) \leq 2\tau_{\text{div}}(a)F_R(a), \quad C_p \geq \sum_{p/4 < a < p/2} \frac{T_{4a-p}(a)}{2\tau_{\text{div}}(a)}.$$

In particular

$$\boxed{C_p > 0 \iff \sum_{p/4 < a < p/2} T_{4a-p}(a) > 0.} \quad (14b)$$

Thus a proof can first force one original p -coloured pair and then apply its exact gcd inverse. It need not establish a termwise lower bound for the Möbius summands. Equation (13) remains necessary when claiming the exact state count.

There is a corresponding finite character formula. For $G = (\mathbb{Z}/R\mathbb{Z})^\times$, let $D_N(\chi) = \sum_{d \mid N} \chi(d)$. Orthogonality gives

$$T_R(N) = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \chi(-1) |1 + \chi(p)|^2 |D_N(\chi)|^2. \quad (14)$$

Indeed the two factors $1 + \chi(p)$ come from the original two p -choices, and the relation $b + c = 0$ supplies $\chi(-1)$ and the conjugate. The elementary orthogonality rule follows by multiplying a nontrivial character sum by a group element at which its value is not one. For $|z| = 1$,

$$\left| \sum_{j=0}^e z^j \right|^2 - \left| \sum_{j=0}^{e-1} z^j \right|^2 = \sum_{j=-e}^e z^j.$$

Applying (12) primewise thus gives the exact original-box expression

$$F_R(N) = \frac{1}{2|G|} \sum_{\chi \in \widehat{G}} \chi(-1) |1 + \chi(p)|^2 \prod_{q^e \parallel N} \left(\sum_{j=-e}^e \chi(q)^j \right). \quad (15)$$

This is a signed sum; no absolute-value or positive-definiteness surrogate is used.

Corollary 7.2 (The source character is annihilated). *At a negative square source, the Jacobi character $\chi_R(\cdot) = (\cdot/R)$ has $\chi_R(p) = -1$. Its entire contribution to (14)–(15) is zero.*

Proof. Theorem 2.1 gives $\chi_R(p) = \chi_p(a) = -1$, so $|1 + \chi_R(p)|^2 = 0$. $\square$

More explicitly, at a negative source the centered Jacobi factor is $\prod_{\ell^e \parallel a} \sum_{j=-e}^e \chi_R(\ell)^j = \chi_p(a) = -1$. Its separate canonical E and M terms are $-1/|G|$ and $+1/|G|$, respectively. They cancel in the joint expression. Thus the precise quadratic character used to force new factors in the source expansion does not itself supply a positive term of the combined count. This identifies a failed proposed implication, not a proof that no stronger arithmetic estimate of (13) can exist.

8. THE RECIPROCAL SOURCE CARRIES E THROUGH A SUM GATE

For $D \geq 3$, $D \equiv 3 \pmod{4}$, put

$$B_D = (pD + 1)/4.$$

No hypothesis $\gcd(p, D) = 1$ is required here: $\gcd(B_D, pD) = 1$ automatically. For $w \mid B_D^2$ satisfying $D \mid w + B_D$, normalize

$$(h, r, s) = (\gcd(B_D, w)^2/w, w/\gcd(B_D, w), B_D/\gcd(B_D, w)), \quad \lambda = (r + s)/D.$$

Retain the input order, and swap r, s to make $r < s$. Equality is impossible: coprimality would give $r = s = 1$, and D cannot divide 2. Set

$$x = hr\lambda, \quad y = hs\lambda, \quad z = pB_D. \quad (16)$$

Theorem 8.1 (Reciprocal sum-gate return). *Equation (16) is an original first-half exterior solution. Its marking is*

$$R' = 4x - p > 0, \quad a' = x, \quad u' = hr^2, \quad (h, r, s_{\text{original}}, \kappa) = (h, r, \lambda, s).$$

The complete inverse is

$$D = (4u' + 1)/R' = (r + \kappa)/s_{\text{original}}, \quad B_D = hr\kappa, \quad w = hr^2 \text{ or } h\kappa^2. \quad (17)$$

Thus every original first-half E state has exactly two reciprocal orientations.

Proof. Since $4hrs = pD + 1$ and $r + s = D\lambda$, direct cancellation in (16) proves the reciprocal identity. Also $R's = pr + \lambda > 0$, and $\gcd(r, \lambda) = 1$ because D is a unit modulo rs . Hence the original E normalization applies. For the upper bound, $x < p/2$ is equivalent to $pD(s - r) > r + s$. Its positive left-minus-right difference is

$$(4hrs - 1)(s - r) - (r + s) = 2s\{2hr(s - r) - 1\} > 0.$$

The same identity gives $x > p/4$. Conversely, an original E state satisfies

$$R'\kappa = pr + s_{\text{original}}, \quad (4u' + 1)/R' = (r + \kappa)/s_{\text{original}}, \quad pD + 1 = 4hr\kappa.$$

The middle equality follows by multiplying by $R's_{\text{original}}$ and using $4hrs_{\text{original}} = p + R'$. Since $\gcd(r, \kappa) = 1$, normalization recovers exactly the two complementary divisors in (17). Only one ordering has $r < \kappa$. $\square$

The dual master equation and its swap are inherited from [2, §§1–4]. Their use is a coordinate comparison, not an assertion of reciprocal-source occupancy.

Every original first-half E state occurs with

$$D \leq \left\lfloor \frac{p^2 + 6p + 13}{12} \right\rfloor. \quad (18)$$

Indeed $u \leq a^2$ gives $D \leq (p^2 + 4)/(4R) + p/2 + R/4$, decreasing for $3 \leq R < p$, and its value at 3 is the bound. Therefore the half-count of all reciprocal sum-gate states in (18) equals the complete E count. The two orientations and the original distinguished denominator are not suppressed in the algorithm.

The reciprocal source has $\chi_p(B_D) = 1$. The same reciprocity proof as before gives $(w/D) = \chi_p(w)$, so every reciprocal hit still requires $\chi_p(w) = -1$. Nonresidue factors can be larger than p in this source: they are not automatically vertices of the original graph.

9. WHAT SIMULTANEOUS SOURCE-FACTOR OVERLAP ACTUALLY PROVES

Assume $p \equiv 1 \pmod{8}$ and $0 < D < 3p$, $D \equiv 3 \pmod{4}$. Define $A_D = (p + D)/4$, $B_D = (pD + 1)/4$.

Theorem 9.1 (Common nonresidue overlap is exactly the endpoint lane).

$$\gcd(A_D, B_D) = \gcd(A_D, (D^2 - 1)/4). \quad (19)$$

Any nonresidue prime ℓ dividing this gcd satisfies

$$\ell \equiv 3 \pmod{4}, \quad \ell \mid p + 1.$$

Conversely, if $p + 1$ has such a prime factor, some $D < p$ gives a common nonresidue prime of both source integers. Hence existence of a common nonresidue overlap in the complete stated D -range is equivalent to this inherited endpoint test.

Proof. Subtract DA_D from B_D to prove (19). A common odd prime gives $D^2 \equiv 1 \pmod{\ell}$ and $p \equiv -D \pmod{\ell}$. If $D \equiv -1$, then $p \equiv 1$ and reciprocity makes ℓ a residue modulo p . Therefore a nonresidue common prime must have $D \equiv 1$, $p \equiv -1$ and $\ell \equiv 3 \pmod{4}$. The prime 2 is a residue and cannot be the exceptional prime. Conversely choose the least $\ell \equiv 3 \pmod{4}$ dividing $p + 1$. The odd integer $(p + 1)/2 \equiv 1 \pmod{4}$ contains an even positive number of such prime occurrences, so $\ell^2 \leq (p + 1)/2$. Thus $D = 2\ell + 1 < p$, $D \equiv 3 \pmod{4}$, and both sources are divisible by ℓ . Reciprocity proves $(\ell/p) = -1$. The endpoint's original E state is $R = \ell$, $a = u = (p + \ell)/4$, $h = a$, $r = s = 1$, $\kappa = (p + 1)/\ell$; this explicitly returns the solution. $\square$

Corollary 9.2 (No common-word reciprocal repair beyond the endpoint). *If $p + 1$ has no prime factor $3 \pmod{4}$, every divisor of $\gcd(A_D, B_D)^2$ has character $+1$ modulo p . Therefore the common original word domain contains no direct E hit and no reciprocal sum-gate hit. When $\chi_p(A_D) = -1$, an original M word in that common domain cannot become a reciprocal sum-gate hit either. A successful reciprocal repair must leave the common word, rather than only reinterpreting its channel.*

Proof. The overlap theorem makes every prime factor of the gcd a quadratic residue. Multiply their characters. Both direct E and reciprocal hits require character -1 by their proved character laws. $\square$

This theorem does not assert that a square-source-restricted D-range always contains the converse D. It identifies why a common-factor coupling cannot, by itself, go beyond the older endpoint lane.

10. A COMPLETE TWO-VERTEX OBSTRUCTION TO THE IMMEDIATE DUAL REPAIR

At $p = 12889$, let $q = 43, 61$ and $D = \sigma_q q t^2 < 3p$, with positive odd t . The complete t-lists are $1, 3, \dots, 29$ and $1, 3, \dots, 13$ respectively. Turn 4 already certified that all 22 original sources have both E/M gates empty, using 246 original divisors [4, §8]. The present certificate additionally factors all 22 reciprocal sources and exhausts their 600 original divisors. Every reciprocal sum gate is empty. Thus 44 numerical source boxes and 66 labelled gate families are empty here. No least-example assertion is made.

For the canonical two reciprocal sources,

$$B_{43} = 127 \cdot 1091, \quad B_{183} = 2^3 \cdot 73709,$$

all prime factors are residues modulo 12889. They add no nonresidue factor edge. Thus the canonical two-cycle $43 \leftrightarrow 61$ remains closed when only these two canonical reciprocal-factor edges are appended. This is NOT a closed component of the full square-source graph, which already has exits.

The original count character and reciprocal total character are opposite: $\chi_p(A_D) = -1$, $\chi_p(B_D) = 1$. Nevertheless all three required gate counts vanish at every displayed D. A reciprocal nonresidue factor alone is insufficient: for example $B_{387} = 37 \cdot 33703$ has nonresidue factors but no sum-gate hit. Some such factors exceed p and cannot be substituted for original graph vertices.

The stronger endpoint test also misses this prime:

$$(p + 1)/2 = 6445 = 5 \cdot 1289,$$

with both factors $1 \pmod{4}$. The overlap theorem therefore forbids common nonresidue source-factor overlap for EVERY D in its interval, not just for the 22 tests. The serialized finite record retains each original factor exponent, candidate divisor, centered vector, fine residue, character and empty/hit observation.

11. AN ACTUAL NON-SQUARE EXCHANGE CROSSES THE CHANNELS

The obstruction has a precise surviving arithmetic map. At the same prime,

$$R = 31, \quad a = 3230 = 2 \cdot 5 \cdot 17 \cdot 19,$$

there are the two original states

$$M : (u, h, r, s, \lambda) = (25, 1, 5, 646, 21),$$

$$E : (u, h, r, s, \kappa) = (85, 85, 1, 38, 417).$$

They give

$$\frac{4}{12889} = \frac{1}{3230} + \frac{1}{174852174} + \frac{1}{1353345} = \frac{1}{3230} + \frac{1}{1346910} + \frac{1}{456850605}. \quad (20)$$

The actual ratio is $u_E/u_M = 17/5 \equiv p^{-1} \pmod{31}$ and has quadratic character -1 . The centered vectors are $(-1, 1, -1, -1)$ and $(-1, 0, 0, -1)$; the shift $(0, -1, 1, 0)$ lies in both original bounds. This is exactly the map (6).

It is reached by a genuine reciprocal factor: the $q=61$, $t=11$ reciprocal source has the complete factorization

$$B_{22143} = 2 \cdot 19 \cdot 31 \cdot 37 \cdot 1637,$$

with nonresidue factors 31 and 37 below p . Selecting 31 gives the original canonical source in (20), which has both channels. This is one verified path, not a theorem that a reciprocal factor must eventually find such an exchange.

For a smaller normalization regression, $p = 193$, $R = 7$, $a = 50$ has four original states but twenty ordered p -coloured pairs. Thus $T_R(a)/2 = 10 \neq 4$. The exact M/E colour counts are $B_0 = 4$, $B_1 = 6$, so the matrix (11) has negative difference eigenvalue. Formula (12), not raw pair counting or a positive-Gram assertion, returns four.

12. THE DECISION FOR TURN 6 AND THE REPLAY BOUNDARY

Three precise limitations have now been identified. The existing unchanged-word and square-ratio transports cannot cross the channels on a negative source. The classical reciprocal source does not rescue every local failure, even after all 22 square ports of the displayed two vertices are retained. Requiring a common nonresidue prime of the two sources collapses to the already known endpoint lane. None is a universal disproof of ES or of all possible channel interactions.

The valid common source (7), its $2\tau_{\text{div}}(h)$ fibres, Möbius transform (13), and character formula (15) give the exact complete-shell object for Turn 6. The quadratic character used in factor expansion is killed in the combined expression; the remaining signed terms must be controlled from actual arithmetic. A positive lower bound for (13) is NOT supplied here. It is the next proof obligation, not another independently postulated positive form.

The shipped standard-library verifier defaults to all first-half shells at primes $p \equiv 1 \pmod{4}$ through 1000, checks exact pair fibres and the Möbius identity, checks the complete reciprocal atlas through $p=97$, and exhausts the 44 displayed source boxes. Its separate implementation shares no main-code import. It uses direct divisor tests, literal quadratic-residue sets and separate factorization. Actual run status, bounds, source hashes and output comparisons are recorded in `verification.json`. No finite replay establishes the universal inequality, no independent mathematical review or Lean build is claimed, and no new overall ES verification range is asserted.

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- [4] The Clankers (2026), *Turn 4: mixed square-source finite reduction and universal six-vertex expansion*, source `es_turn4_square_expansion_20260918/core.tex`, PR19 revision 4b6bb0f187fa44bb62fc8c51e996f05dfc90e2d7. Only the original source, carry and displayed 12889 fixture are reused here; the large cofactor enumeration is not rerun or assumed to force channel occupancy.